Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software Engineer with 2+ years building **reliable cloud infrastructure workflows**, **Kubernetes-adjacent services**, and **observability-oriented backend tooling** using Python, AWS services, SQL, and CI/CD automation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **scalable backend workflows** on **Azure SQL/PostgreSQL** for 500+ users, reducing latency from **1.2s to 350ms** through query-path optimization, indexing strategy, and service-level tuning.
- Implemented **resilience-focused validation pipelines** for 5K+ monthly jobs with reconciliation checks, improving incident triage, operational reliability, and deployment confidence for production infrastructure-adjacent services.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Developed **Python simulation pipelines** processing 100K+ trajectory logs, enabling systematic performance evaluation and rapid iteration on compute-heavy workflows with reproducible execution and debugging artifacts.
- Documented validation outcomes and fault patterns, strengthening post-mortem quality and improving cross-team handoffs for productionizing research workflows into stable platform capabilities.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **API and ETL pipelines** with OOP, AWS Glue, and S3, improving response time by **30%** across distributed integration paths and cloud-based data services.
- Maintained **20+ unit/integration suites** with Jenkins, PyTest, and Mockito across three OS environments, improving pipeline efficiency by **68%** and reducing technical debt by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Built **REST API and WebSocket services** supporting eight workflows and 2M+ synchronized records, improving service communication reliability and fault visibility across multi-component production operations.
- Automated deployments and testing with **Docker** and Jenkins, reducing deployment time by **50%** while strengthening repeatability and rollback safety in fast-paced release environments.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Collaborated across product and engineering teams to ship **Flask + React** features serving 10K+ daily requests, balancing speed, reliability, and maintainable implementation quality.
- Added **CI/CD safeguards** and concurrent processing controls, reducing turnaround time by **60%** while improving deployment stability and measurable service behavior under peak operational load.

Skills

Languages: **Python**, Java, SQL, JavaScript/TypeScript, C++
Infrastructure & Cloud: **AWS**, Kubernetes concepts (EKS/ECS familiarity), Docker, VPC/SG/ALB/Route53 awareness, API Gateway concepts
Platform Engineering: Linux workflow, distributed systems basics, service reliability, cloud-native operational patterns
Observability & Quality: Datadog/Splunk familiarity, unit/integration testing, Jenkins, PyTest, CI/CD, incident documentation
Data Services: PostgreSQL, MySQL, MongoDB, Azure SQL, ETL orchestration with AWS Glue and S3

Projects

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*
- Built **Kubernetes-backed platform workflows** for 50K+ users and 1M+ transactions, improving scalability, service communication reliability, and deployment consistency under high-throughput distributed operational conditions.

Chat2SQL Code | *Java, TypeScript, SQL, Maven, Docker*
- Developed a multi-database backend engine spanning 10+ databases with **schema-aware validation**, structured execution flows, and maintainable service boundaries for reliable API-driven query processing.

HMIS Live | *MERN, ABDM APIs, Twilio*
- Designed **monitoring-backed API workflows** supporting 100+ daily appointments, improving retrieval latency by **2x** and strengthening fault diagnosis across production services requiring secure, dependable operations.